

Exploring Distributed Synthesis: In-Progress Findings from Guided Tours of Scholarly Knowledge Synthesis Work Practices with A Distributed Lens

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ABSTRACT

Synthesis — the construction of a new conceptual whole that advances understanding — is critical for research progress. Its current practice is also extremely laborious and costly. In this paper, we explore what fresh ideas CSCW might have to offer for making effective synthesis less painful and more commonplace. In particular, we draw on CSCW ideas around distributed sensemaking and integrated crowdsourcing to explore the idea of distributed synthesis: systems for sharing and reusing intermediate synthesis materials such as annotations and notes to kickstart and accelerate the synthesis process. We report in-progress findings from our analysis of in-depth guided tours of scholarly synthesis work practices, to bridge the gap between distributed sensemaking and integrated crowdsourcing (which has not yet been studied in the setting of scholarly synthesis) and scholarly synthesis workflows (which have not yet been analyzed from a distributed lens).

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *Interactive systems and tools*; **Empirical studies in collaborative and social computing**.

KEYWORDS

synthesis, sensemaking, knowledge reuse, distributed collaboration

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1 INTRODUCTION

Synthesis — the construction of a new conceptual whole that advances understanding — is critical for research progress [26, 36, 43]. Synthesis can take a variety of forms, such as theories, integrative reviews, or design space models: the common thread is their powerful benefits for inspiring progress, by clearly revealing what is

unknown, and potential paths to new discoveries and innovation [5, 40]. Our work here responds to the specific observation that synthesis work is, by many accounts, extremely arduous [8, 15, 23] and cost-intensive [7, 8, 28, 32, 38]. Audits of synthesis quality suggest that the level of infeasibility may be translating into significant deficits in synthesis quality in published research outputs, such as literature reviews in doctoral dissertations [6, 16, 25] and published papers [2–4, 11, 27, 43].

What might CSCW research have to offer for this problem? And what might new insights about CSCW might be gained through deep reflection on this problem context? We think there may be some fresh ideas to draw from existing CSCW work around **distributed sensemaking** [10, 14, 17, 21, 22, 30] — structured hand-offs of intermediate sensemaking products such as annotations and schemas for others to use as starting points for their own sensemaking — and **integrated crowdsourcing** — seamlessly integrating ordinarily laborious secondary tasks that are valuable for others (e.g., semantic judgments) into a primary task that people are already intrinsically motivated to do [19, 39, 44]. As two concrete examples, Kittur et al [21] designed a system that allowed consumers to draw on categories/tags and clippings from previous consumers searching for similar products, to kickstart and structure their sensemaking and foraging process while doing research to purchase consumer products like digital cameras; and Siangliu-lue et al [39] built a system that drew information about semantic similarity between ideas from the intrinsically motivating act of clustering ideas (on virtual post-its) on a virtual whiteboard, which was then used to construct a semantic space that could be used for mapping a solution space in collective brainstorming. These design patterns demonstrate models for self-sustaining, virtuous cycles of collective intelligence.

Drawing on these ideas, we are curious about the possibility of constructing **distributed synthesis systems**. Could we construct self-sustainable systems for synthesis that draw intermediate synthesis materials — annotations, tags, notes — from primary tasks of the research lifecycle, such as reading and note-taking, manuscript preparation, or peer review, and feed into a distributed system that helped other researchers more quickly find and synthesize relevant ideas for their problems from the literature? Could this distributed approach help to address persistent problems in existing efforts to incentive and sustain contributions of synthesis materials, from traditional forms, such as review papers, to newer forms, such as semantic "micropublishing" of networks of granular claims and their context [24, 35]. As a concrete example of this authorship bottleneck, the popular Mark2Cure project — a citizen science approach



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to extracting gene-drug-disease relationships from biomedical abstracts [42] — had to shut down due to inability to sustain enough participation to curate enough knowledge to uncover clues for identifying potential treatment strategies for NGLY1-deficiency [13].

The problem that motivates our study is the disconnect between the ideas of distributed sensemaking and integrated crowdsourcing with the specific setting of scholarly synthesis. The design patterns of distributed sensemaking and integrated crowdsourcing are well-studied in some domains, such as doing travel planning or researching consumer products [10, 21], design brainstorming [39], intelligence analysis [30], and learning [20], but have not yet been directly applied to the setting of scholarly synthesis. On the flip side, we do have rich descriptions of scholarly practices, but without a particular focus on potential connections between individual/collaborative practice and distributed synthesis. For example, past work has articulated the various types of information artifacts that are produced during synthesis work, such as contextual information [5, 29], hypothesis projections [5], in-source annotations, per-source summaries, and cross-source syntheses [34]. From a process-perspective, we have extensions of individual sensemaking models [33] to describe the synthesis process as similarly non-linear and iterative [12, 34] and interleaved with multiple information behaviors, including information foraging, information retrieval, and sensemaking [31]. However, these descriptions have not yet been systematically analyzed with particular attention to questions of retrieval and reuse of information artifacts, either by oneself or others.

To design systems for integrating contributions to distributed synthesis systems, we need to understand more deeply questions like *"What information artifacts produced and used during individual synthesis might be useful for sharing with others in a distributed manner?"* and *"What information practices and tooling affordances in individual synthesis might serve as points of integration into distributed synthesis systems?"*.

2 METHODS

To address these questions, we conducted guided tour studies [9, 41] to investigate synthesis practices from a distributed perspective. For this study, we recruited 21 researchers. Our participants were quite diverse in terms of disciplines and topics, including computer science, cognitive science, law, mathematics, economics, and business, and more (see Table 1 for more details); the unifying factor was the prominence of synthesis (vs. engineering/implementation) in their research projects, and such as mapping out new design spaces to push the boundaries of some design problem (P01, P02, P05, P10, P12, P13, P15, P16), critical reviews of and theory-building from prior literature (P04, P06, P09, P11, P14). Additionally, given our focus on discovering potentially new affordances/practices we might integrate distributed synthesis systems into, we recruited participants in part from user communities of newer or niche tools such as RoamResearch¹, a personal hypertext notebook, in addition to recruiting participants from within academia/industry who use traditional tools and workflows, such as Google Docs. Table 1 shows

the tools and research areas of the participants we have analyzed so far.

The participants took part remotely via video meetings. Participants first shared with us the specific research directions for which they are conducting synthesis work, which grounded the guided tour of their synthesis practices. Then, for the guided tour portion, they shared their screens and walked through their synthesis workflows, first for their individual synthesis work, and then for collaborative synthesis work (if any). Throughout, we asked followup and probing questions about changes to their workflows/systems, sources of structure and the forms of artifacts, and remaining pain points. Each interview lasted approximately 1-1.5 hours. After transcribing the recordings into text-based transcripts, the first author performed open coding on these interview transcripts and captured screenshots of the participants' shared screens displaying their work processes, and iterated on the codes through discussion with the second author. The open coding was focused on the participants' behavior regarding information artifacts, aiming to identify the relationship between information reuse at the individual level and the collaborative level.

3 PRELIMINARY RESULTS

To date, we have analyzed 16 participants (see Table 1). So far, we have observed more instances of reusing information artifacts for individual synthesis work, compared to sharing for collaborative synthesis work. However, we have sufficient data to notice emerging themes around the differences in challenges associated with reusing information artifacts in individual synthesis work compared to those encountered when sharing synthesis information artifacts with others in collaborative synthesis work.

3.1 Reuse in individual synthesis

A few participants reported that contextual information was important for successfully reusing previously encountered or processed information during individual synthesis work. For example, P4 confirmed whether his current understanding aligns with previously highlighted texts by using contextual information, saying *"You don't want to just take out the highlight. You want to double check your things, your understanding of this unit. So, like few lines above and below, just to make sure that you get it right."* P13 also described how margin notes support his understanding of original quotes that were highlighted in the past, saying, *"(margin notes make it) so much easier for me to jump to this quote here and say, cool, I remember exactly why I highlighted this is because I wanted to make it clear that we're not focusing on education."*

We also observed how both under-structured and overly structured notetaking systems had a negative influence on participants' effectiveness in reusing information artifacts in their individual synthesis work. A common consequence of these systems was an increase in difficulties for participants when attempting to retrieve the desired information artifacts. On the low structure end, P4 mentioned that his messy file management system prevented him from finding a specific paper, saying, *"I have a papers folder on my computer, that is a mess."* Similarly, P14 had too many related documents which made his system less structured and hard

¹<https://roamresearch.com/>

Table 1: Participants' Tools Used for Synthesis and Their Research Directions

ID	Tools Used	Synthesis Research Direction/Questions
P01	Bear ²	novel software interface that can support cognition and creativity
P02	Apple Notes	video games as a medium for communication
P03	Logseq ³	political and economic liberty from a Buddhist perspective
P04	PDF readers, Blog, Ulysses ⁴	tacit knowledge and mental models
P05	RoamResearch ⁵	types of graph visualization in note-taking applications
P06	Logseq	laws in Central American countries
P07	Bear/Figma	visualization of infinite sets of different sizes in the mathematical sense.
P08	Miro	capability models for product development
P09	Blog, Notion	longevity research, economic history of the Soviet Union
P10	Tinderbox ⁶	new areas where Wikipedia could incorporate transclusion
P11	New Things Under the Sun	economics of innovation, including gender and problem choice in science
P12	Notion	software capable of deconstructing PDF documents
P13	PDF/Print Outs	comprehensive tools for designers, crowdsourcing
P14	Google Doc	strategies for developing new psychological theories
P15	Figma	novel NFTs for watch faces
P16	Workflowy/Miro/GoogleDoc	future of spreadsheets, search engine optimization

to locate the paper he wanted, complaining, “*There are too many related documents. I don’t know which is which.*”

On the other hand, P11 had elaborate tagging and naming systems, such as long reading lists with tags and abstracts. Despite this, they reported that this system was still insufficient, saying, “*I haven’t found a great solution that like I could put a lot of work into trying to tag them.*”. Elaborating on this, P11 complained, “*They get buried. I’ll never find them again.*”. Similarly, P12 set too many hierarchical levels in his management system. As a result, his system became overly structured and hindered his retrieval process. He elaborated, “*Like I was constantly reshuffling where these drafts were and like burying them under multiple levels of hierarchy and like my Notions.*”

3.2 Sharing in collaborative synthesis

In contrast to what was observed when participants discussed their workflows for individual synthesis, successful (and, more commonly, unsuccessful) sharing in collaborative synthesis work was more frequently associated with broader shared background knowledge and skills with collaborators (or the lack thereof). For example, P8 mentioned that it is challenging for him to share his thoughts using his visualized maps with people who don’t possess similar skillsets. P8 gained the skills of synthesizing at multiple levels which is different from his collaborators, mentioning “*Miro supercharged my learning journey into being able to create multiple levels compared to people in amplitude.*” Adding to this, P8 explained that the gaps were produced because his collaborators don’t practice the same tools and methods, explaining, “*Because they don’t practice this all day. This was very difficult for them.*” Therefore, the artifacts that P8 created were difficult to share with his collaborators, saying, “*There’s a huge problem that one thing could be associated with multiple things. Like that was like, a problem for them.*” P10 also described how his information artifacts are hard to share to somebody do not have the same level of experiences with the same type of information artifacts, saying, “*How do I take this sort of, you know, rather*

boring looking outline, which I know quite well, because I’ve been working on this on and off for years but share it with somebody else.”

4 DISCUSSION

4.1 Initial Implication: Reuse by self vs. others

While our findings are still in progress, a theme of a contrast between reuse by self vs. others in the synthesis process is emerging as interesting from both a design perspective as well as a potential contribution to CSCW more generally. As discussed in our introduction, past research on scholarly synthesis work has identified the non-linear process of reusing information artifacts as typical; however, our findings suggest that the notion of “reuse” in this sense may be substantially different from the notion of reuse in more classical CSCW work on organizational memory and knowledge sharing, which typically refers to sharing information between different individuals (reuse by others) [1]. To clarify this distinction in our work, we use the term “reuse” here to refer to reuse by self (i.e., for individual synthesis work), and “sharing” to refer to reuse by others (i.e., in collaborative synthesis work).

Our in-progress findings suggest that reusing and sharing are distinct scenarios within the synthesis process, each with different material requirements to be successful. The details of required contextual information differ for reusing and sharing: in the reusing process, synthesis failures often occur due to a lack of cues to retrieve previous lines of thinking. This is consistent with some prior studies of reuse in knowledge work [18, 37]. In contrast, failures of sharing primarily stem from differences in background knowledge and skills among individuals. This is also consistent with themes from prior studies of sharing knowledge in collaborative work settings [1].

Yet, we could frame reusing and sharing as facing a common challenge in current synthesis practices: the need to retrieve appropriate *contextual* information alongside primary material for synthesis: in the individual case, it is things like cues to previous thinking,

and in the collaborative case, it is related information about structure/conventions and background knowledge. Further, we reason that information (along with their contextual information) that cannot be retrieved for individual reuse also cannot be shared with others. Thus, one interesting point to ponder is whether the information artifact management structures that enable retrieval of contextual information may also be repurposed or integrated into systems for retrieving contextual information to accompany sharing of information artifacts with others, either in collaborative settings, or in more distributed synthesis networks. We are curious whether these information management structures could be further enhanced with technological mechanisms, such as (semi-)automated tagging, enhanced search or question-answering systems (e.g., with large language models), or contextual computing and modeling.

4.2 Next Steps

We plan to augment our current findings by more deeply comparing reusing and sharing behavior in participants' workflows of synthesis. First, we want to identify what type of information artifacts and participants' behaviors support and differentiate successful reusing and sharing. Second, we want to investigate at which stage of participants' workflows synthesis is more capable of being translated to a distributed way and integrated with distributed synthesis systems. Our diverse sample also presents the need (and opportunity) to understand how the challenges and requirements of reuse/sharing may vary systematically within the sample; for example, we are noticing some variations in synthesis orientation (focused on relatively well-structured problems, vs. open-ended, ill-structured problems). Finally, we want to aggregate our study findings to inform a roadmap for successful synthesis systems.

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REFERENCES

- [1] Mark S. Ackerman, Juri Dachtera, Volkmar Pipek, and Volker Wulf. 2013. Sharing Knowledge and Expertise: The CSCW View of Knowledge Management. *Computer Supported Cooperative Work (CSCW)* 22, 4-6 (Aug. 2013), 531–573. <https://doi.org/10.1007/s10606-013-9192-8>
- [2] Adrienne Alton-Lee. 1998. A Troubleshooter's Checklist for Prospective Authors Derived from Reviewers' Critical Feedback. *Teaching and Teacher Education* 14, 8 (1998), 887–90.
- [3] Mats Alvesson and Jörgen Sandberg. 2011. Generating research questions through problematization. *Academy of management review* 36, 2 (2011), 247–271. Publisher: Academy of Management Briarcliff Manor, NY.
- [4] Sheetal Bhurke, Andrew Cook, Anna Tallant, Amanda Young, Elaine Williams, and James Raftery. 2015. Using systematic reviews to inform NIHR HTA trial planning and design: a retrospective cohort. *BMC Medical Research Methodology* 15, 1 (Dec. 2015), 108. <https://doi.org/10.1186/s12874-015-0102-2> 00000.
- [5] Catherine Blake and Wanda Pratt. 2006. Collaborative information synthesis I: A model of information behaviors of scientists in medicine and public health. *Journal of the American Society for Information Science and Technology* 57, 13 (2006), 1740–1749. <https://doi.org/10.1002/asi.20487> 00097_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/asi.20487>
- [6] David N. Boote and Penny Beile. 2005. Scholars Before Researchers: On the Centrality of the Dissertation Literature Review in Research Preparation. *Educational Researcher* 34, 6 (Aug. 2005), 3–15. <https://doi.org/10.3102/0013189X034006003>
- [7] Rohit Borah, Andrew W. Brown, Patrice L. Capers, and Kathryn A. Kaiser. 2017. Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open* 7, 2 (Feb. 2017), e012545. <https://doi.org/10.1136/bmjopen-2016-012545> Publisher: British Medical Journal Publishing Group Section: Health informatics.
- [8] Ann-Margret Ervin. 2008. Motivating authors to update systematic reviews: practical strategies from a behavioural science perspective. *Paediatric and perinatal epidemiology* 22, 0 1 (Jan. 2008), 33–37. <https://doi.org/10.1111/j.1365-3016.2007.00910.x>
- [9] Michele C. Everett and Margaret S. Barrett. 2012. "Guided tour": a method for deepening the relational quality in narrative research. *Qualitative Research Journal* 12, 1 (April 2012), 32–46. <https://doi.org/10.1108/1443988121122271400000>
- [10] Kristie Fisher, Scott Counts, and Aniket Kittur. 2012. Distributed Sensemaking: Improving Sensemaking by Leveraging the Efforts of Previous Users. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12)*. ACM, New York, NY, USA, 247–256. <https://doi.org/10.1145/2207676.2207711>
- [11] Padhraig S. Fleming, Jadbinder Seehra, Argy Polychronopoulou, Zbys Fedorowicz, and Nikolaos Pandis. 2013. Cochrane and non-Cochrane systematic reviews in leading orthodontic journals: a quality paradigm? *European Journal of Orthodontics* 35, 2 (April 2013), 244–248. <https://doi.org/10.1093/ejo/cjs016> Publisher: Oxford Academic.
- [12] Allen Foster. 2004. A nonlinear model of information-seeking behavior. *Journal of the American Society for Information Science and Technology* 55, 3 (2004), 228–237. <https://doi.org/10.1002/asi.10359> 00468_eprint: <https://asistdl.onlinelibrary.wiley.com/doi/pdf/10.1002/asi.10359>
- [13] ginger. 2019. Thank you, Volunteers and Citizen Scientists. <http://mark2cure.org/blog/thank-you-campaign-pause/00000>
- [14] Nitesh Goyal and Susan R. Fussell. 2016. Effects of Sensemaking Translucence on Distributed Collaborative Analysis. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW '16)*. ACM, New York, NY, USA, 288–302. <https://doi.org/10.1145/2818048.2820071>
- [15] Darcy Haag Granello. 2001. Promoting Cognitive Complexity in Graduate Written Work: Using Bloom's Taxonomy as a Pedagogical Tool to Improve Literature Reviews. *Counselor Education and Supervision* 40, 4 (June 2001), 292–307. <https://doi.org/10.1002/j.1556-6978.2001.tb01261.x>
- [16] Allyson Holbrook, Sid Bourke, Terence Lovat, and Kerry Dally. 2004. Investigating PhD thesis examination reports. *International Journal of Educational Research* 41, 2 (Jan. 2004), 98–120. <https://doi.org/10.1016/j.ijer.2005.04.008>
- [17] Ruogu Kang and Sara Kiesler. 2012. Do collaborators' annotations help or hurt asynchronous analysis. In *Proceedings of the ACM 2012 conference on Computer Supported Cooperative Work Companion*. ACM, Seattle Washington USA, 123–126. <https://doi.org/10.1145/2141512.2141558> 00011.
- [18] Alison Kidd. 1994. The marks are on the knowledge worker. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '94)*. Association for Computing Machinery, New York, NY, USA, 186–191. <https://doi.org/10.1145/191666.191740>
- [19] Juho Kim, Phu Tran Nguyen, Sarah Weir, Philip J. Guo, Robert C. Miller, and Krzysztof Z. Gajos. 2014. Crowdsourcing Step-by-step Information Extraction to Enhance Existing How-to Videos. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. ACM, New York, NY, USA, 4017–4026. <https://doi.org/10.1145/2556288.2556986>
- [20] Juho Kim, Phu Tran Nguyen, Sarah Weir, Philip J. Guo, Robert C. Miller, and Krzysztof Z. Gajos. 2014. Crowdsourcing Step-by-Step Information Extraction to Enhance Existing How-to Videos. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. ACM, New York, NY, USA, 4017–4026. <https://doi.org/10.1145/2556288.2556986>
- [21] Aniket Kittur, Andrew M. Peters, Abdigani Diriye, and Michael Bove. 2014. Standing on the Schemas of Giants: Socially Augmented Information Foraging. In *Proceedings of the 17th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '14)*. ACM, New York, NY, USA, 999–1010. <https://doi.org/10.1145/2531602.2531644>
- [22] Aniket Kittur, Andrew M. Peters, Abdigani Diriye, Trupti Telang, and Michael R. Bove. 2013. Costs and Benefits of Structured Information Foraging. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. ACM, New York, NY, USA, 2989–2998. <https://doi.org/10.1145/2470654.2481415>
- [23] Ian A. Knight, Max L. Wilson, David F. Brailsford, and Natasha Milic-Frayling. 2019. Enslaved to the Trapped Data: A Cognitive Work Analysis of Medical Systematic Reviews. In *Proceedings of the 2019 Conference on Human Information Interaction and Retrieval (CHIIR '19)*. ACM, New York, NY, USA, 203–212. <https://doi.org/10.1145/3295750.3298937> event-place: Glasgow, Scotland UK.
- [24] Tobias Kuhn and Michel Dumontier. 2017. Genuine semantic publishing. *Data Science* 1, 1-2 (Jan. 2017), 139–154. <https://doi.org/10.3233/DS-170010>
- [25] Barbara E. Lovitts. 2007. *Making the Implicit Explicit: Creating Performance Expectations for the Dissertation*. Stylus Publishing, Sterling, Va.
- [26] Richard McElreath and Paul E. Smaldino. 2015. Replication, Communication, and the Population Dynamics of Scientific Discovery. *PLOS ONE* 10, 8 (Aug. 2015), e0136088. <https://doi.org/10.1371/journal.pone.0136088> 00107 Publisher: Public Library of Science.
- [27] Jonathon McPhetres, Nihan Albayrak-Aydemir, Ana Barbosa Mendes, Elvina C. Chow, Patricio Gonzalez-Marquez, Erin Loukras, Annika Maus, Aoife O'Mahony,

- Christina Pomareda, Maximilian Primbs, Shalaine Sackman, Conor Smithson, and Kirill Volodko. 2020. *A decade of theory as reflected in Psychological Science (2009–2019)*. Technical Report. PsyArXiv. <https://doi.org/10.31234/osf.io/hs5nx> 00000 type: article.
- [28] Matthew Michelson and Katja Reuter. 2019. The significant cost of systematic reviews and meta-analyses: A call for greater involvement of machine learning to assess the promise of clinical trials. *Contemporary Clinical Trials Communications* 16 (Dec. 2019), 100443. <https://doi.org/10.1016/j.conctc.2019.100443>
- [29] John S Morabito and Joel Chan. 2021. Managing Context during Scholarly Knowledge Synthesis: Process Patterns and System Mechanics. In *Creativity and Cognition (C&C '21)*. Association for Computing Machinery, New York, NY, USA, 1–5. <https://doi.org/10.1145/3450741.3465244> 00000.
- [30] Syavash Nobarany, Mona Haraty, and Brian Fisher. 2012. Facilitating the Reuse Process in Distributed Collaboration: A Distributed Cognition Approach. In *Proceedings of the ACM 2012 Conference on Computer Supported Cooperative Work (CSCW '12)*. ACM, New York, NY, USA, 1223–1232. <https://doi.org/10.1145/2145204.2145388>
- [31] Carole L. Palmer, Lauren C. Tefreau, and Carrie M. Pirmann. 2009. *Scholarly Information Practices in the Online Environment: Themes from the Literature and Implications for Library Service Development*. Technical Report. 59 pages. <http://www.oclc.org/programs/publications/reports/2009-02.pdf>
- [32] Anthony Petrosino. 1999. Lead authors of cochrane reviews: Survey results. *Report to the Campbell Collaboration*. Cambridge, MA: University of Pennsylvania (1999).
- [33] Peter Pirolli and Stuart Card. 2005. The Sensemaking Process and Leverage Points for Analyst Technology as Identified through Cognitive Task Analysis. In *Proceedings of International Conference on Intelligence Analysis*, Vol. 5. 2–4.
- [34] Xin Qian, Katrina Fenlon, Wayne G. Lutters, and Joel Chan. 2020. Opening Up the Black Box of Scholarly Synthesis: Intermediate Products, Processes, and Tools.. In *Proceedings of ASIST 2020*.
- [35] Allen H. Renear and Carole L. Palmer. 2009. Strategic Reading, Ontologies, and the Future of Scientific Publishing. *Science* 325, 5942 (Aug. 2009), 828–832. <https://doi.org/10.1126/science.1157784>
- [36] Anne M. Scheel, Leonid Tiokhin, Peder M. Isager, and Daniël Lakens. 2020. Why Hypothesis Testers Should Spend Less Time Testing Hypotheses. *Perspectives on Psychological Science: A Journal of the Association for Psychological Science* (Dec. 2020), 1745691620966795. <https://doi.org/10.1177/1745691620966795> 00008.
- [37] Moushumi Sharmin, Brian P. Bailey, Cole Coats, and Kevin Hamilton. 2009. Understanding knowledge management practices for early design activity and its implications for reuse. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, Boston MA USA, 2367–2376. <https://doi.org/10.1145/1518701.1519064> 00047.
- [38] Kaveh G. Shojania, Margaret Sampson, Mohammed T. Ansari, Jun Ji, Steve Doucette, and David Moher. 2007. How Quickly Do Systematic Reviews Go Out of Date? A Survival Analysis. *Annals of Internal Medicine* 147, 4 (Aug. 2007), 224. <https://doi.org/10.7326/0003-4819-147-4-200708210-00179>
- [39] Pao Siangliulue, Joel Chan, Steven P. Dow, and Krzysztof Z. Gajos. 2016. Idea-Hound: Improving Large-Scale Collaborative Ideation with Crowd-Powered Real-Time Semantic Modeling. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16)*. ACM, New York, NY, USA, 609–624. <https://doi.org/10.1145/2984511.2984578>
- [40] Kenneth Strike and George Posner. 1983. Types of synthesis and their criteria. In *Knowledge Structure and Use*, S Ward and L Reed (Eds.). Temple University Press, Philadelphia, PA.
- [41] Leslie Thomson. 2015. The guided tour technique in information science: Explained and illustrated. *Proceedings of the Association for Information Science and Technology* 52, 1 (2015), 1–5. <https://doi.org/10.1002/pri.2015.1450520100135>
- [42] Ginger Tsueng, Max Nanis, Jennifer T. Fouquier, Michael Mayers, Benjamin M. Good, and Andrew I. Su. 2019. Applying Citizen Science to Gene, Drug, Disease Relationship Extraction from Biomedical Abstracts. *bioRxiv* (Feb. 2019), 564187. <https://doi.org/10.1101/564187> 00002 Publisher: Cold Spring Harbor Laboratory Section: New Results.
- [43] Iris van Rooij and Giosuè Baggio. 2021. Theory Before the Test: How to Build High-Verisimilitude Explanatory Theories in Psychological Science. *Perspectives on Psychological Science* (Jan. 2021), 1745691620970604. <https://doi.org/10.1177/1745691620970604> 00029 Publisher: SAGE Publications Inc.
- [44] Sarah Weir, Juho Kim, Krzysztof Z. Gajos, and Robert C. Miller. 2015. Learn-ersourcing Subgoal Labels for How-to Videos. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '15)*. ACM, New York, NY, USA, 405–416. <https://doi.org/10.1145/2675133.2675219>